Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Core run I, family 'free flight'. In the early flight, the speed and the direction can change freely. The speed is Mach 0.45 to 0.87 until 18:40, then Mach 0.70 to 0.87, always between 210 and 330 knots indicated air speed. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight.

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 36.83°S and mode 37.05°S, with 95% of probability between 30.44°S and 38.14°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S. The two densities overlap by 64%. This curve is centred 0.7° further north and carries 37% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the same family in run D, the median moves 0.11° north, the 34.5–36.5°S shoulder holds 37% of probability (base 29%), and the two curves overlap by 78%.

The curve pools 8 independent replicates of 7,000,000 particles. Single replicates overlap each other by 46%–86% and their medians span 0.93°. Over all balanced partitions of the 8 replicates the split-half overlap averages only 83.0% (range 75.8% to 90.3%) against the 8-replicate floor of 0.924, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.

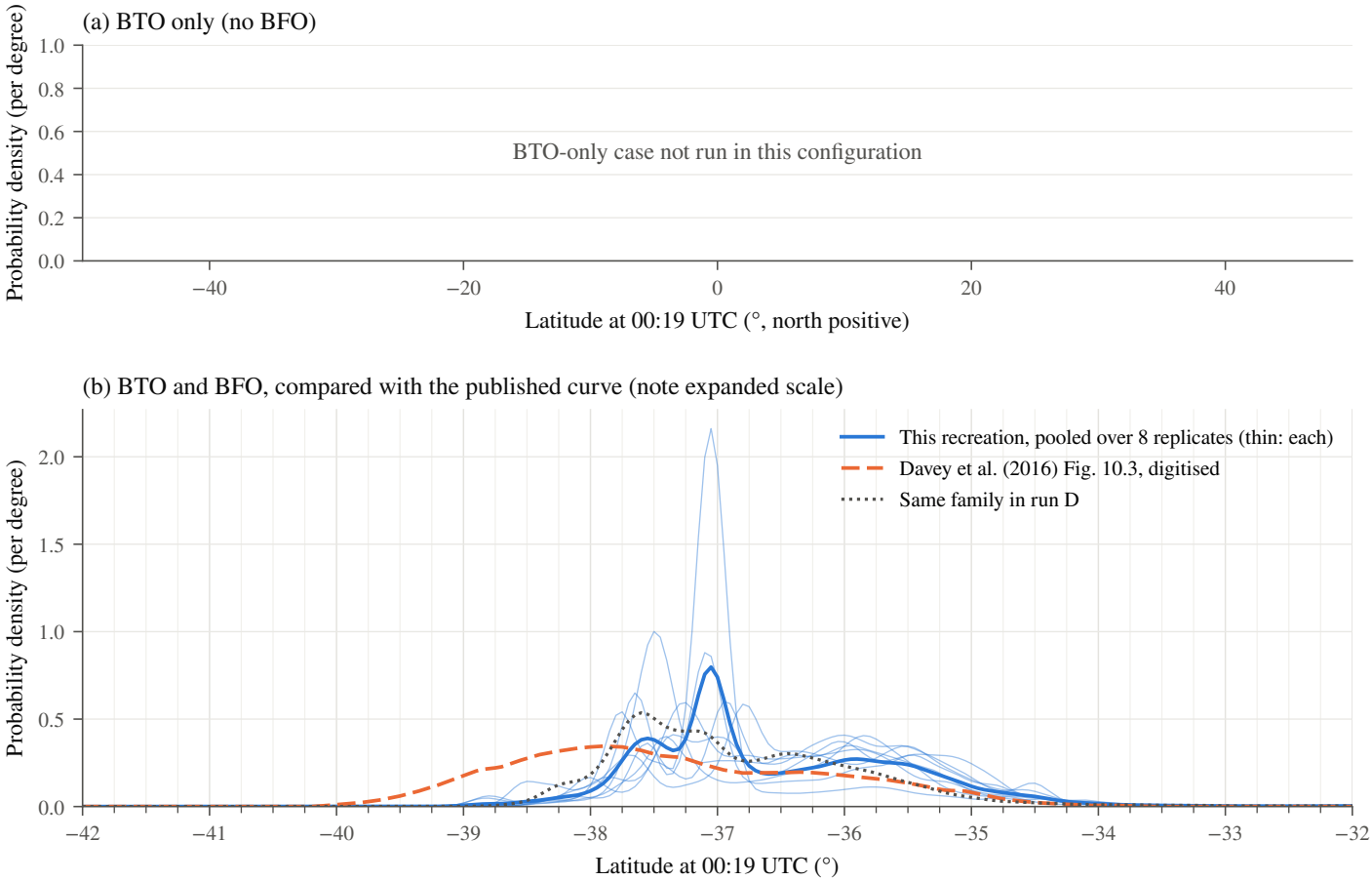


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. (a) BTO only. (b) BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. This run also uses: primary-radar positions and track segments 18:04-18:22 scored as measurements; fuel model internal-v1, two tanks, one-engine flight after the first flame-out (hold-taper drift-down), paths dry before 00:11 rejected. No debris, drift, search or end-of-flight evidence is used. Run conditions as stated at the top of this page.

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Technical: run next-i-free; stack davey2016-inmarsat + no-exhaustion-prior + s1-factor + s2-temperature + s3-internal + s4-ceiling + s5-hard-reject + s6-tanks + reference-snapshots + radar-full + s7-single-engine + s8-hold-taper + inop-flow-fix + speed-envelope-throughout + handoff-lookahead-d + free + server-i-free; binary 1ff4d1d, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -135.78; split-half mean over 35 balanced partitions 0.830 (range 0.758-0.903) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); outputs compacted (tanks32, gzipped hand-offs; final.npy unchanged); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

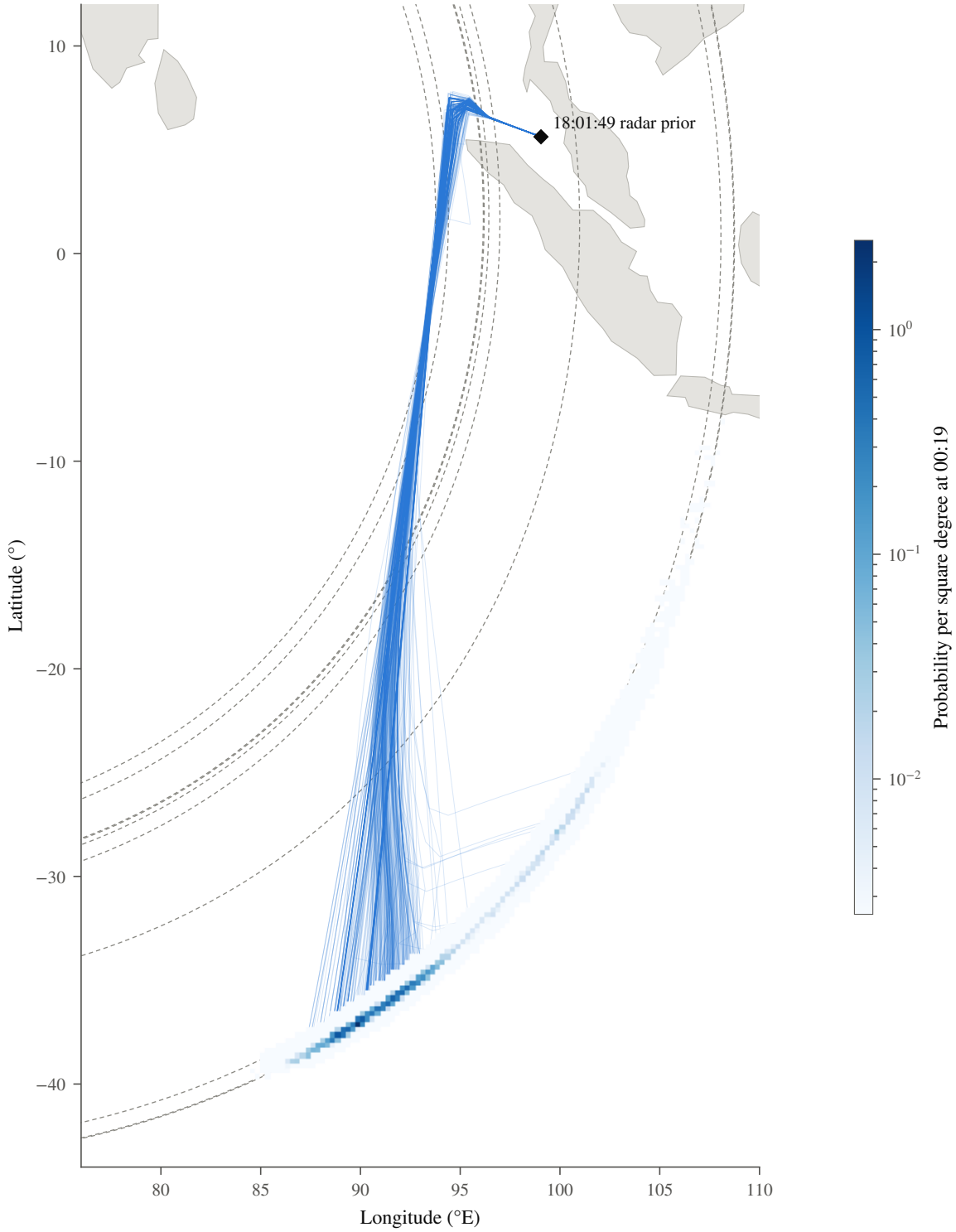
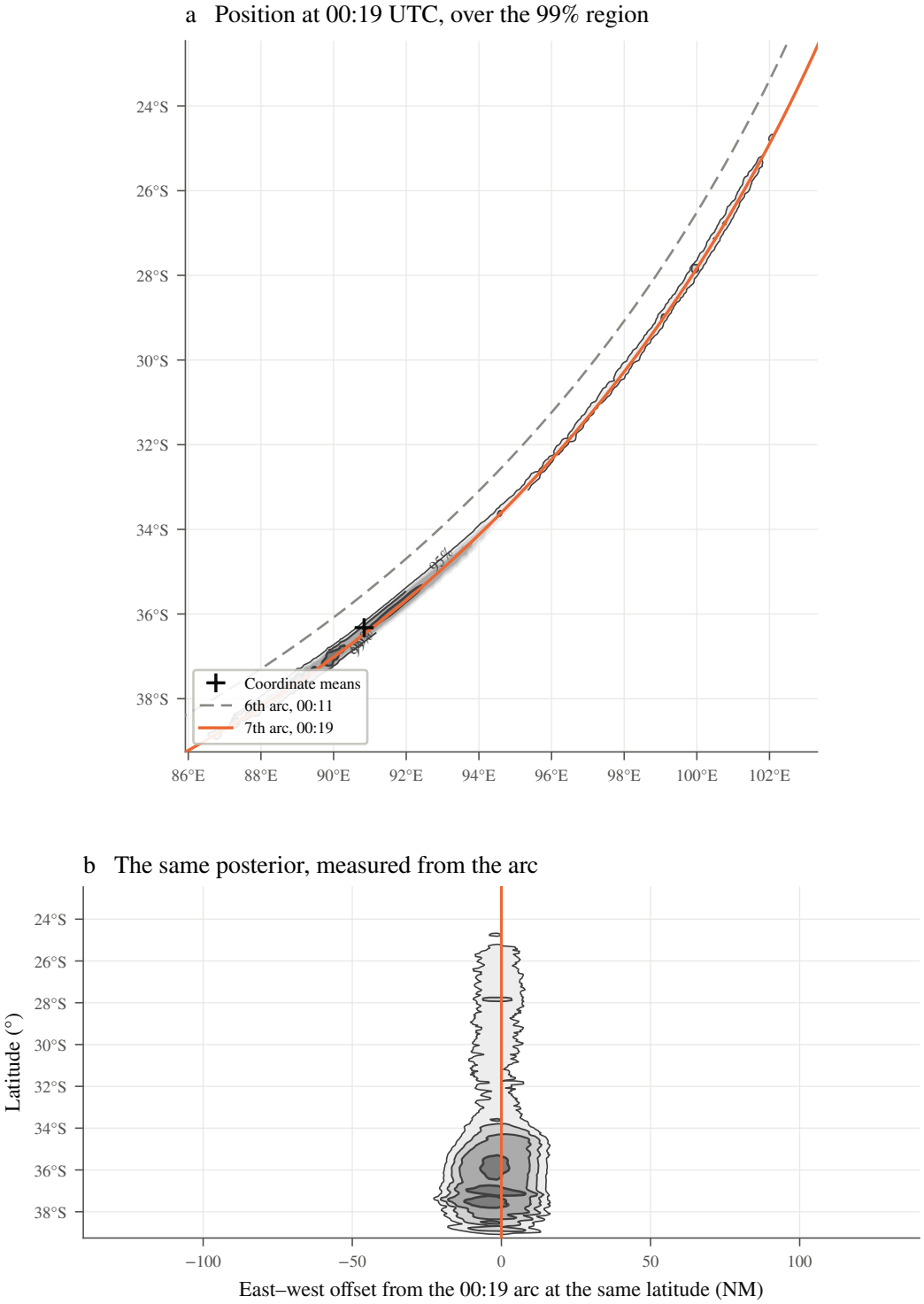


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 8 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

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DIFFERENCES FROM DAVEY

Mach after an acceleration, uniform on: [0.7, 0.87]

Manoeuvre integration step: 5

Position sd: 0.5

Initial Mach set point: -

Prior mean position and track:

BTO noise sd, R600: 63 at 1 epoch

Vertical rate in the BFO model: True

Weather source: era5-wind-temperature.bin

Magnetic declination source: igrf14-declination.bin

Resampling scheme: one filter per autopilot …

Cost-index speed mode:

Fuel and endurance: {'factor_mean': 1.0004, '…

Fuel proposal: endurance (deadline unix …

book values: see the parameter page

POSTERIOR AT 00:19

median 36.83°S, mode 37.05°S

95% 30.44–38.14°S

replicates 8, split-half 0.734

largest single draw 0.00% of weight

99% region 18,627 NM²

95% region 8,970 NM²

90% region 5,964 NM²

50% region 1,422 NM²

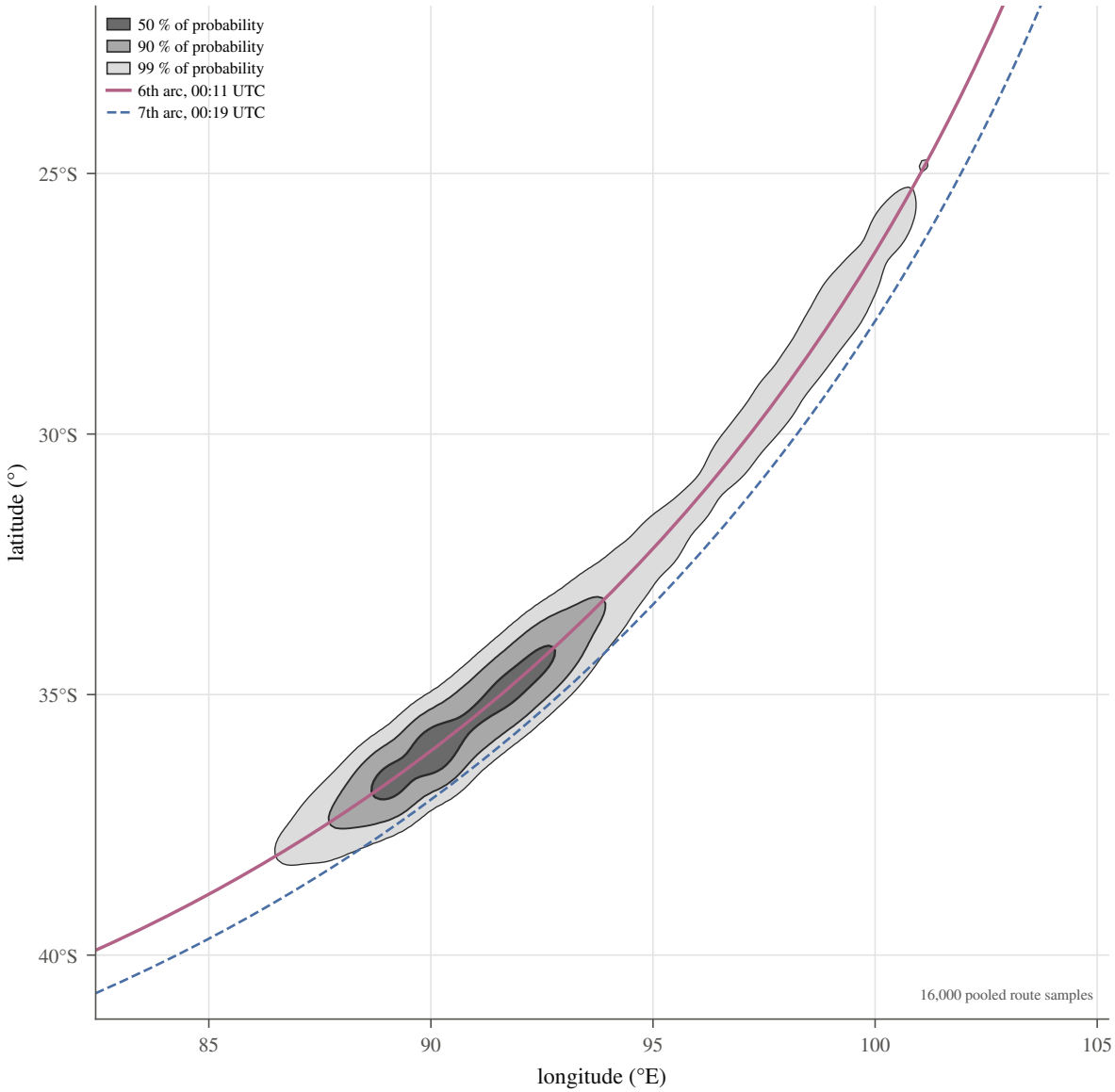
Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 18,627, 95% 8,970, 90% 5,964, 50% 1,422. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 17° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

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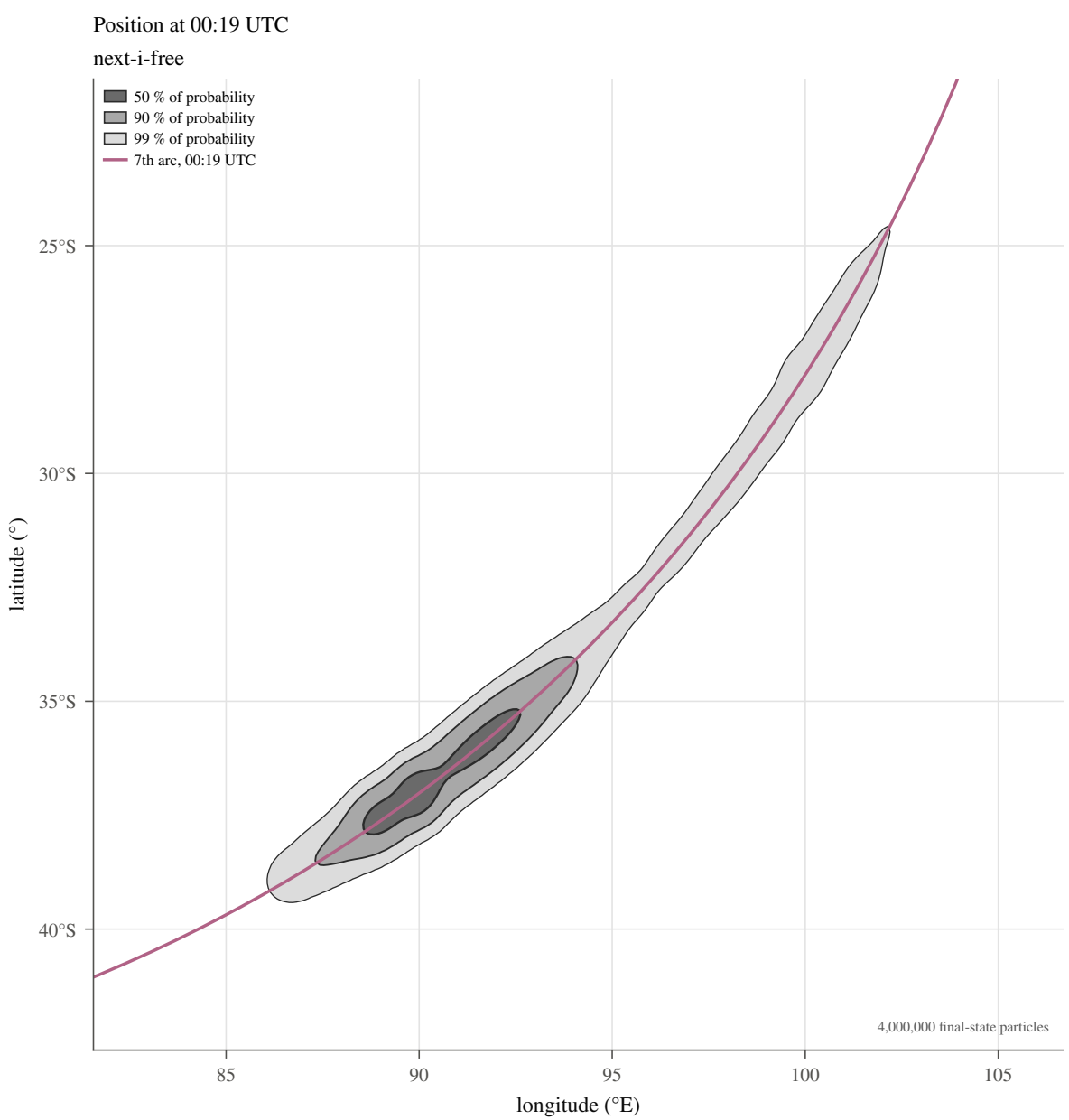
Position at 00:11 UTC

next-i-free



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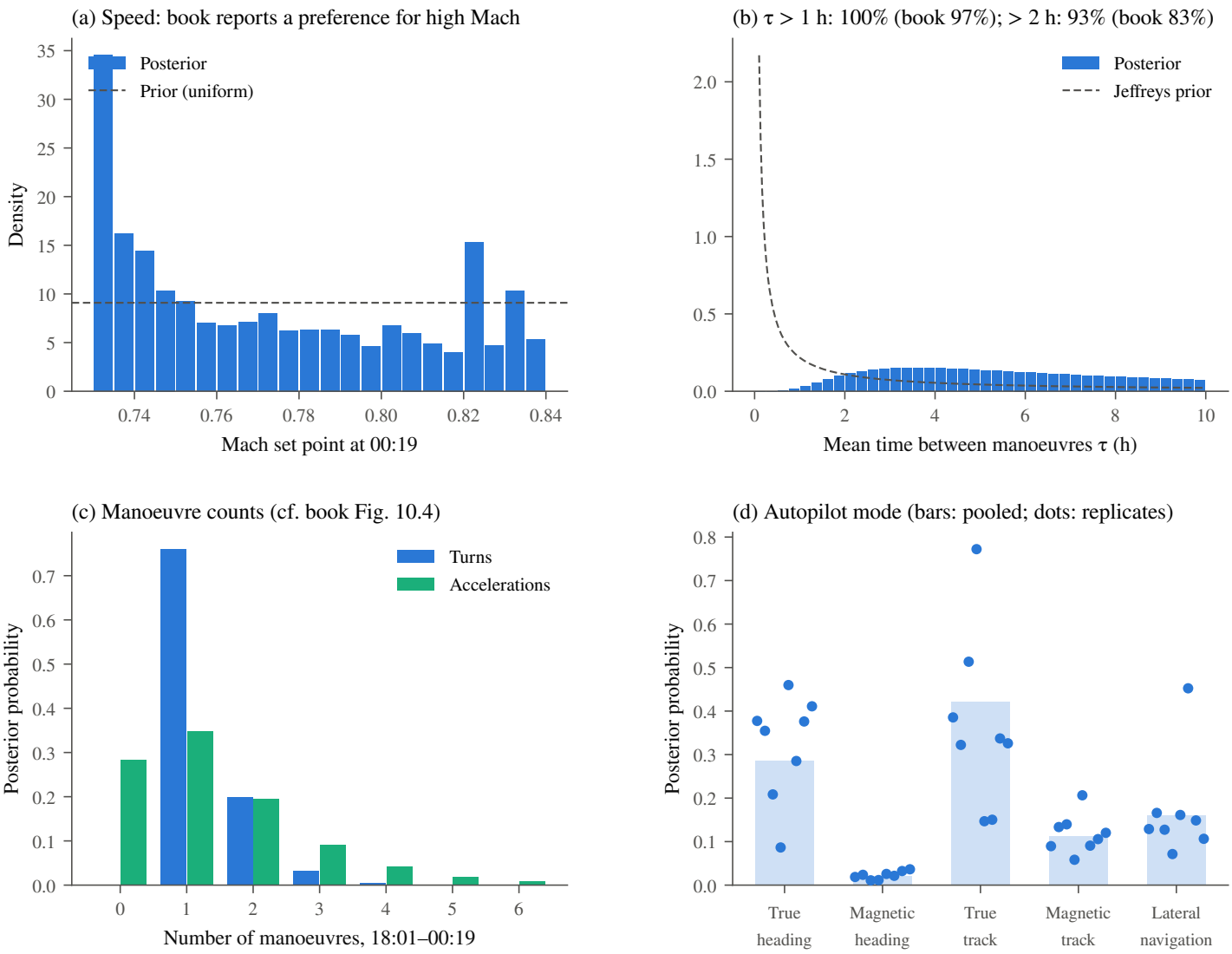
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Posterior behaviour compared with the book



	This recreation	Davey et al. (2016)
Median latitude	36.83°S	37.53°S
Mode	37.05°S	37.90°S
95% interval	30.44–38.14°S	34.90–39.38°S
95% width	7.70°	4.47°
Overlap with published pdf	64% (replicates 46%–66%)	—
Median Mach at 00:19	0.766	upper part of 0.73–0.84
P(north of equator), BTO only	not run	northern mode present

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

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Method, convergence and reproduction

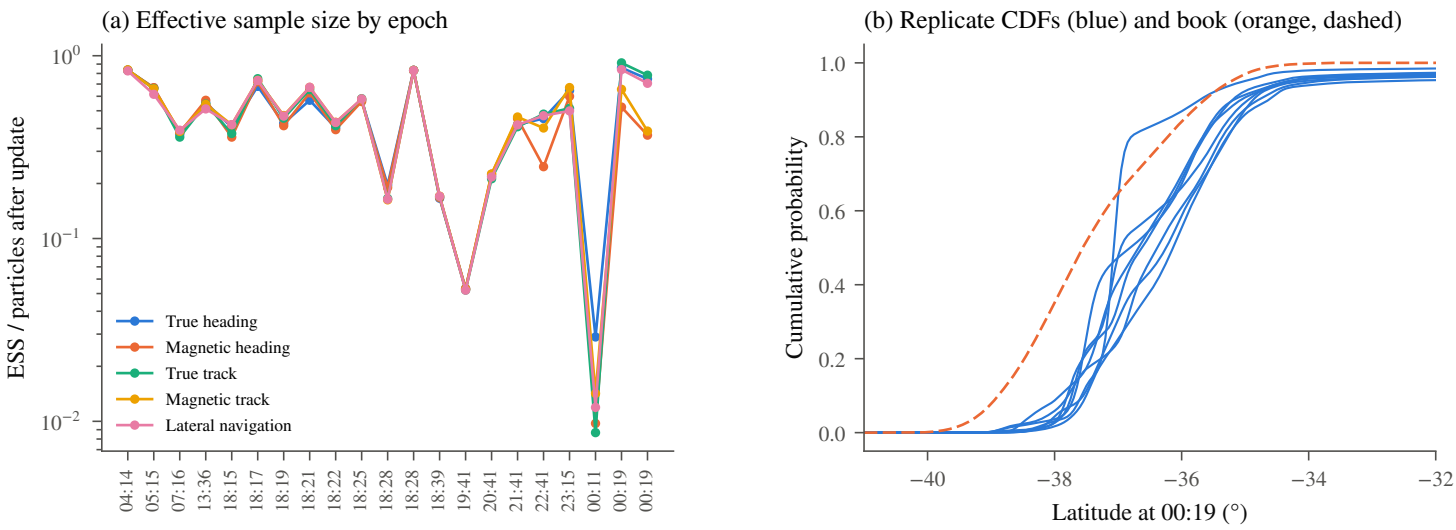


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 8 independent replicates; the largest difference between any two is 0.493 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 µs); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-i-free; particles per mode 1,000,000, 500,000, 2,500,000, 500,000, 2,500,000 (order as in Figure 3d), 7,000,000 per replicate; BTO+BFO seeds 1, 2, 3, 4, 5, 6, 7, 8; BTO-only seeds ; code revision 1ff4d1d; 88 threads; total runtime 100.1 min; peak memory 30,347 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-free', varying cas_envelope_kt (not declared) to [210.0, 330.0]; mach_range [0.73, 0.84] to [0.7, 0.87]; replicates 4 to 8.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

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Parameters and model choices against Davey et al. (2016)

43 parameters and choices. 30 match the published model; 13 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

	Parameter	This run	Davey et al. (2016)
	Dynamics		
	Mach OU reversion rate (1/s)	0.01058	0.01058
	Mach OU noise strength (1/s)	2.05e-07	2.05e-07
	Control-angle OU reversion rate (1/s)	0.009792	0.009792
	Control-angle OU noise strength (rad^2/s)	4.074e-08	4.074e-08
	Wind-error OU reversion rate (1/s)	0.001087	0.001087
	Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
	Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
▶	Mach after an acceleration, uniform on (Mach)	[0.7, 0.87]	[0.73, 0.84]
	Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
	Altitude step (ft)	1000	1000
	Bank angle in turns (deg)	15	15
	Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
	Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
	Mean time to lateral-navigation reversion (s)	31162.2	31162.2
	Cruise integration step (s)	10	10
▶	Manoeuvre integration step (s)	5	1
	Prior		
	Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶	Position standard deviation (NM)	0.5	0.4
	Track standard deviation (deg)	1	1
	BFO bias prior mean (Hz)	150	150
	BFO bias prior standard deviation (Hz)	25	25
▶	Initial Mach set point	-	[0.82, 0.03]
▶	Prior mean position and track	-	not tabulated
	Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,…	equal over the five modes
	Measurement		
	BTO fixed offset (us)	495679	495679
	BTO noise sd, R1200 (us)	29 at 7 epochs	29
	BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶	BTO noise sd, R600 (us)	63 at 1 epoch	62
	BFO noise sd (Hz)	7 at 11 epochs	7
	BFO bias treatment	constant per trajectory, Kalman-margina…	constant per trajectory, Kalman-marginalised
	Prior		
	BFO bias drift rate (Hz^2/s)	0	0
	Measurement		
	BTO bias treatment	-	fixed per-flight T_channel only; no bias st…
▶	Vertical rate in the BFO model	True	not modelled
	00:19 BFO	-	not used; believed unreliable
	Model choice		
▶	Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000…
▶	Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶	Resampling scheme	one filter per autopilot mode combined…	randomised branching: weight >= eta branche…
	Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre…
	Turn extent (deg)	-	uniform on -180 to +180
▶	Cost-index speed mode	-	implemented but unused in most experiments;…
▶	Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0.…	infinite fuel; constraints applied to the p…
	Sampler		
▶	Fuel proposal	endurance (deadline unix 1394237459, pr…	n/a (no fuel model)
	Model choice		
	Observation cutoff	m0019b	00:19 (BTO only at 00:19)

▶ marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

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